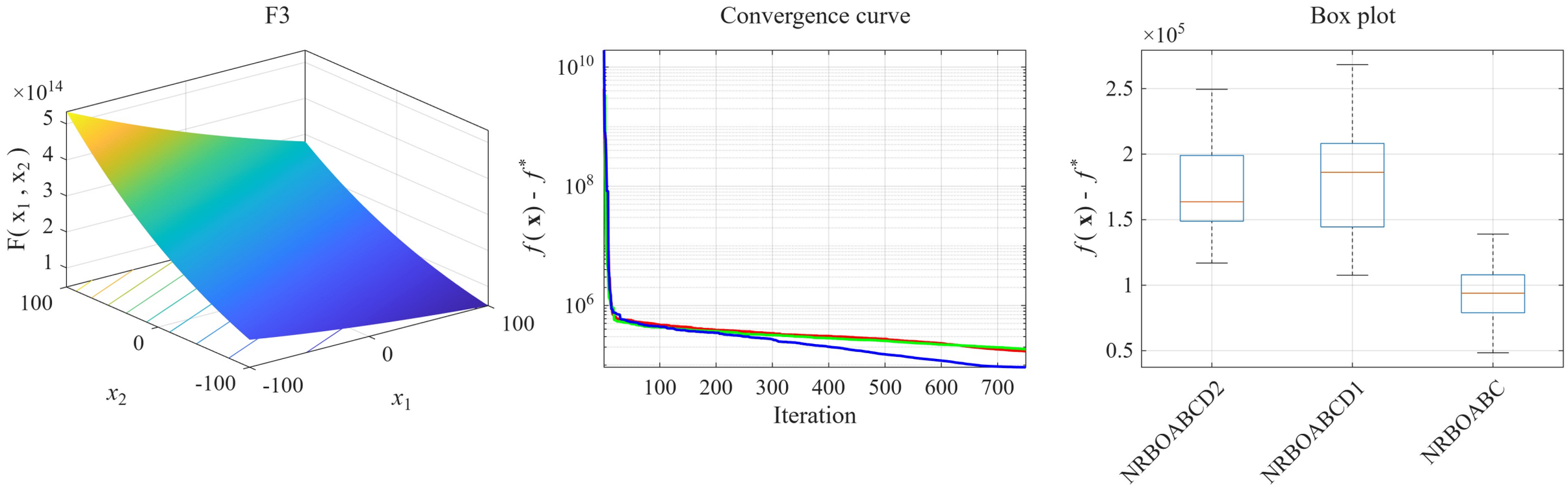

F3
×10^14
F(x1 , x2)
5
4
3
2
1
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^10
10^8
10^6
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^5
2.5
2
1.5
1
0.5
f(x) - f*
NRBOABCD2
NRBOABCD1
NRBOABC
NRBOABCD2
NRBOABCD1
NRBOABC